

1. What is a Database? What is DBMS?

Follow-up Questions:

- Why do we use a DBMS instead of storing data in files?
 - Give some examples of DBMS software.
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2. What is the difference between DBMS and RDBMS?

Follow-up Questions:

- What is a relational database?
 - Give examples of RDBMS.
 - Which one have you used in your projects?
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3. What is a Primary Key? What are the characteristics of a Primary Key?

Follow-up Questions:

- Can a table have more than one primary key?
 - Can a primary key contain NULL values?
 - What is a Composite Primary Key?
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4. What is the difference between a Primary Key and a Foreign Key?

Follow-up Questions:

- Why is a Foreign Key used?
 - Can a Foreign Key contain duplicate values?
 - Can a Foreign Key contain NULL values?
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5. Explain the different types of SQL commands.

Follow-up Questions:

- What is the difference between DDL, DML, DCL, TCL, and DQL?
 - Give two commands for each category.
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6. What is Normalization?

Follow-up Questions:

- Why is normalization important?
 - What problems does it solve?
 - Explain 1NF, 2NF, and 3NF with simple examples.
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7. What are Joins in SQL?

Follow-up Questions:

- Explain INNER JOIN.
 - Explain LEFT JOIN.
 - Explain RIGHT JOIN.
 - Explain FULL OUTER JOIN.
 - When would you use each join?
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8. What is the difference between DELETE, TRUNCATE, and DROP?

Follow-up Questions:

- Which command can be rolled back?
 - Which command removes the table structure?
 - Which command is faster and why?
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9. What is an Index?

Follow-up Questions:

- Why do we create indexes?
 - Can too many indexes affect performance?
 - Where are indexes commonly used?
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10. What are ACID Properties?

Follow-up Questions:

- Explain Atomicity.
- Explain Consistency.
- Explain Isolation.
- Explain Durability.
- Why are ACID properties important in transactions?

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